

Project Plan

Project Title

Alien Invaders Game

1. Project Description

This project is a simple arcade-style game created using python's turtle module and guizero for GUI elements. The player controls a spaceship at the bottom of the screen that can move left and right and shoot bullets upward to destroy incoming alien ships. The game keeps track of the score using a GUI display, and the player loses if an alien reaches the bottom.

The purpose of the project is to practice programming concepts such as loops, functions, classes, and collision detection, while creating a visually interactive game.

2. Objectives of the Project

The goals of this project are:

- To create a fun shooting game using turtle graphics
- To implement player movement using keyboard controls
- To use functions and classes to organize game objects
- To track score and display it using guizero
- To implement collision detection between bullets and aliens

3. Concepts That Will Be Used

The project will apply the following Python concepts:

- Variables and data types
- Lists
- Functions

- Classes and objects
- For loops / While loops
- Event handling for keyboard input
- Turtle graphics for drawing and animation
- guizero for score display and GUI elements

4. Project Features

The program will include the following features:

- A game screen showing the player spaceship and aliens
- Multiple alien enemies descending from the top
- The spaceship can move left and right
- The spaceship can shoot bullets upward
- Collision detection between bullets and aliens
- Score tracking displayed in a guizero window
- Game over condition when an alien collides with the spaceship

5. Basic Project Structure

Step 1

Import required libraries

```
import turtle
```

```
import random
```

```
from guizero import App, Text
```

Step 2

Create the game screen using Turtle and set the background

Step 3

Create the player spaceship and alien objects

Step 4

Store aliens in a list for easy management

Step 5

Use a loop to move aliens downward and bullets upward

Step 6

Detect collisions and update the score

Step 7

Display game over when an alien reaches the bottom

6. Example Functions

Possible functions that may be used:

- `create.aliens()` – generate alien objects
- `draw_spaceship()` – display the player's spaceship
- `fire_bullet()` – shoot bullets from the spaceship
- `move.aliens()` – control alien movement
- `check_collision()` – detect bullet-alien collisions
- `update_score()` – update the guizero score display

7. Possible Challenges

Some possible challenges may include:

- Moving multiple aliens and bullets simultaneously in Turtle
- Accurate collision detection between bullets and aliens
- Keeping the player's spaceship responsive to keyboard input
- Integrating guizero GUI elements with Turtle animation

8. Expected Output

The program will display a visually interactive game where the spaceship shoots bullets at descending aliens. The score updates in a separate GUI panel, and the game ends if any alien collides with the player.